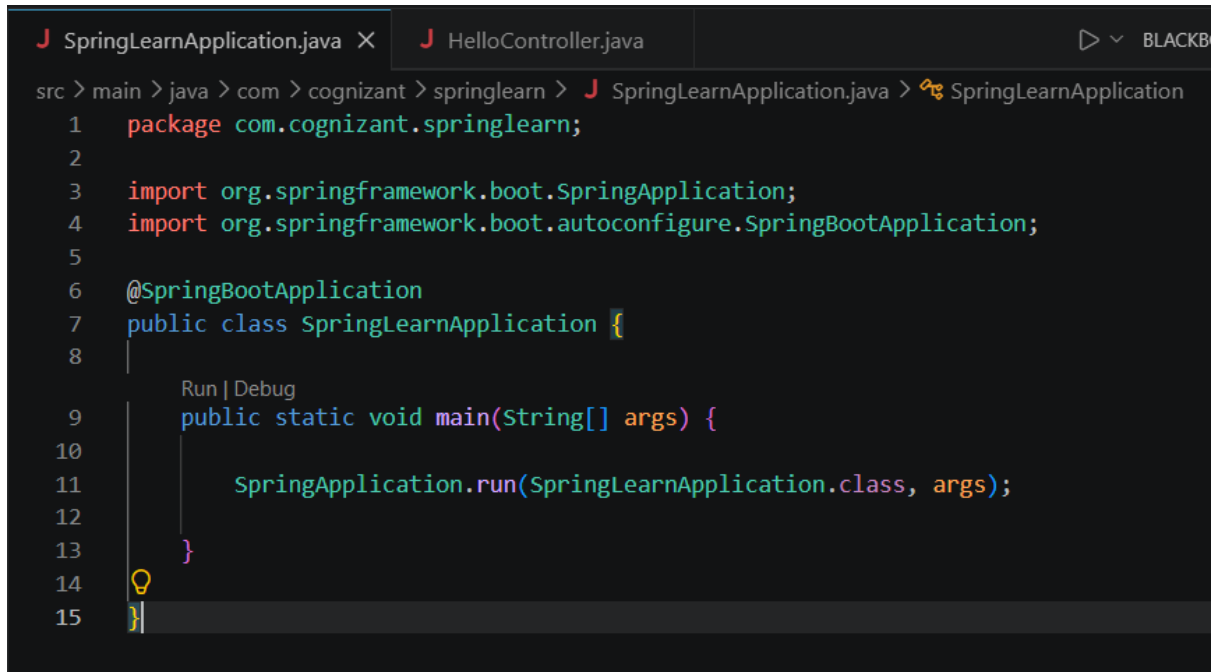
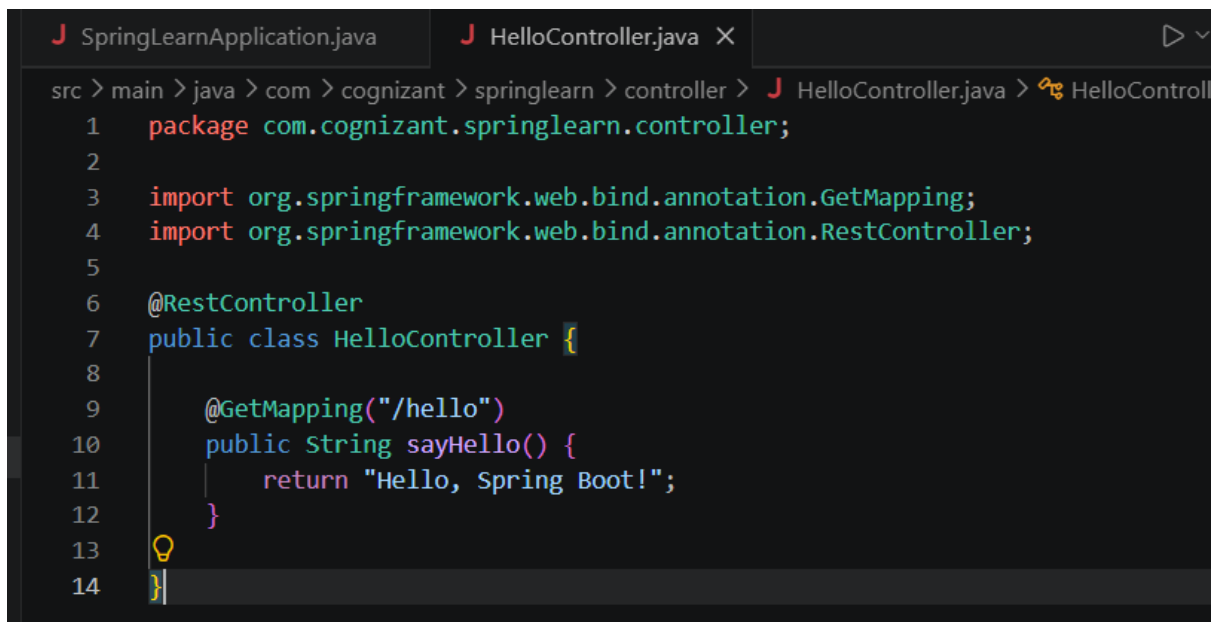


Exercise 1: Create a Spring Web Project using Maven

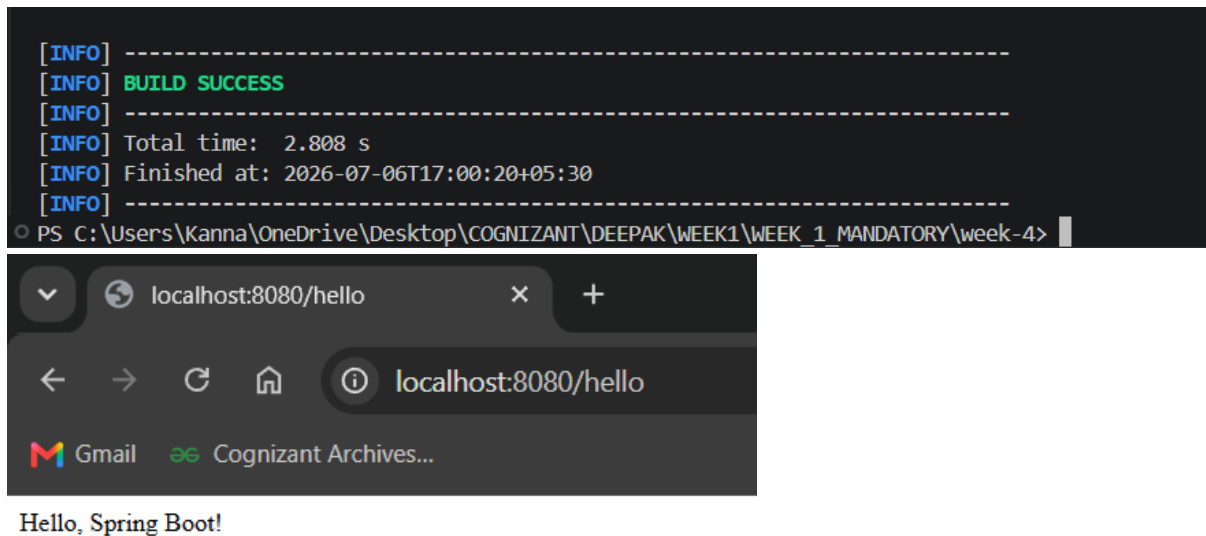


```
J SpringLearnApplication.java X J HelloController.java
src > main > java > com > cognizant > springlearn > J SpringLearnApplication.java > SpringLearnApplication
1 package com.cognizant.springlearn;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6 @SpringBootApplication
7 public class SpringLearnApplication {
8
9     public static void main(String[] args) {
10
11         SpringApplication.run(SpringLearnApplication.class, args);
12     }
13 }
14
15
```



```
J SpringLearnApplication.java J HelloController.java X
src > main > java > com > cognizant > springlearn > controller > J HelloController.java > HelloController
1 package com.cognizant.springlearn.controller;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.RestController;
5
6 @RestController
7 public class HelloController {
8
9     @GetMapping("/hello")
10     public String sayHello() {
11         return "Hello, Spring Boot!";
12     }
13 }
14
```

RESULT :



Exercise 2: Spring Core – Load Country from Spring Configuration XML

```
package com.cognizant.springlearn;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class SpringLearnApplication {

    public static void main(String[] args) {

        displayCountry();

    }

    public static void displayCountry() {

        System.out.println("Loading Spring context from country.xml...");

        ApplicationContext context =
            new ClassPathXmlApplicationContext("country.xml");

        Country country =
            context.getBean("country", Country.class);

        System.out.println("Country : " + country);

    }
}
```

```
J SpringLearnApplication.java 1 J Country.java X application.properties country.xml
src > main > java > com > cognizant > springlearn > J Country.java > Country > setCode(String)
1 package com.cognizant.springlearn;
2
3 public class Country {
4
5     private String code;
6     private String name;
7
8     public Country() {
9         System.out.println("Inside Country Constructor.");
10    }
11
12    public String getCode() {
13        System.out.println("Inside getCode()");
14        return code;
15    }
16
17    public void setCode(String code) {
18        System.out.println("Inside setCode()");
19        this.code = code;
20    }
21
22    public String getName() {
23        System.out.println("Inside getName()");
24        return name;
25    }
26
27    public void setName(String name) {
28        System.out.println("Inside setName()");
29        this.name = name;
30    }
31
32    @Override
33    public String toString() {
34        return "Country{code='" + code + "', name='" + name + "'}";
35    }
36 }
```

```
src > main > resources > country.xml
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4     xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5     xsi:schemaLocation="
6         http://www.springframework.org/schema/beans
7         https://www.springframework.org/schema/beans/spring-beans.xsd">
8
9     <bean id="country"
10         class="com.cognizant.springlearn.Country">
11
12         <property name="code" value="IN"/>
13
14         <property name="name" value="India"/>
15
16     </bean>
17
18 </beans>
```

RESULT :

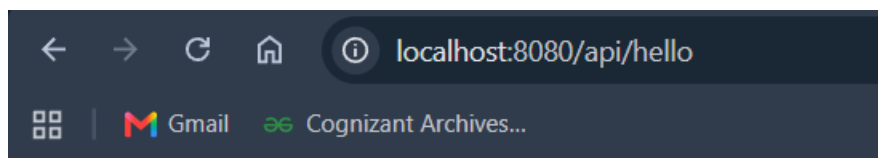
```
[INFO] -----< com.cognizant:spring-learn >-----
[INFO] Building spring-learn 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- exec:3.6.3:java (default-cli) @ spring-learn ---
Loading Spring context from country.xml...
Inside Country Constructor.
Inside setCode()
Inside setName()
Country : Country{code='IN', name='India'}
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.284 s
[INFO] Finished at: 2026-07-06T17:10:11+05:30
[INFO] -----
PS C:\Users\Kanna\OneDrive\Desktop\COGNIZANT\DEEPAK\WEEK1\WEEK 1 MANDATORY\week-4>
```

Exercise 3: Hello World RESTful Web Service

```
J HelloRestController.java X J SpringLearnApplication.java
src > main > java > com > cognizant > springlearn > controller > J HelloRestController.java
1 package com.cognizant.springlearn.controller;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.RestController;
5
6 @RestController
7 public class HelloRestController {
8
9     @GetMapping("/api/hello")
10    public String hello() {
11        return "Hello from REST service!";
12    }
13
14 }
```

```
J HelloRestController.java  J SpringLearnApplication.java X
src > main > java > com > cognizant > springlearn > J SpringLearnApplication.java > SpringLearn
1  package com.cognizant.springlearn;
2
3  import org.springframework.boot.SpringApplication;
4  import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6  @SpringBootApplication
7  public class SpringLearnApplication {
8
9      Run | Debug
10     public static void main(String[] args) {
11         SpringApplication.run(SpringLearnApplication.class, args);
12     }
13 }
```

RESULT :



Hello from REST service!

```
StatusCode      : 200
StatusDescription :
Content         : Hello from REST service!
RawContent      : HTTP/1.1 200
                  Keep-Alive: timeout=60
                  Connection: keep-alive
                  Content-Length: 24
                  Content-Type: text/plain; charset=UTF-8
                  Date: Mon, 06 Jul 2026 11:50:17 GMT
```

Exercise 4 REST - Country Web Service

```
J HelloRestController.java X J SpringLearnApplication.java country.xml
src > main > java > com > cognizant > springlearn > controller > J HelloRestController.java
1 package com.cognizant.springlearn.controller;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.RestController;
5
6 @RestController
7 public class HelloRestController {
8
9     @GetMapping("/api/hello")
10    public String hello() {
11        return "Hello from REST service!";
12    }
13
14 }
```

```
J HelloRestController.java J SpringLearnApplication.java country.xml X
src > main > resources > country.xml
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5       xsi:schemaLocation="
6         http://www.springframework.org/schema/beans
7         https://www.springframework.org/schema/beans/spring-beans.xsd"
8
9     <bean id="country"
10          class="com.cognizant.springlearn.Country">
11
12         <property name="code" value="IN"/>
13         <property name="name" value="India"/>
14
15     </bean>
16
17 </beans>
```

RESULT :

```
StatusCode      : 200
StatusDescription :
Content         : {"code":"IN","name":"India"}
RawContent      : HTTP/1.1 200
                  Keep-Alive: timeout=60
                  Connection: keep-alive
                  Content-Length: 28
                  Content-Type: application/json
                  Date: Mon, 06 Jul 2026 12:00:05 GMT

                  {"code":"IN","name":"India"}
Forms           : {}
Headers         : {[Keep-Alive, timeout=60], [Connection, keep-alive], [Content-Length, 28],
                  [Content-Type, application/json]...}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 28
```

Pretty-print ☐

```
{"code":"IN","name":"India"}
```

Exercise 5 REST - Get country based on country code

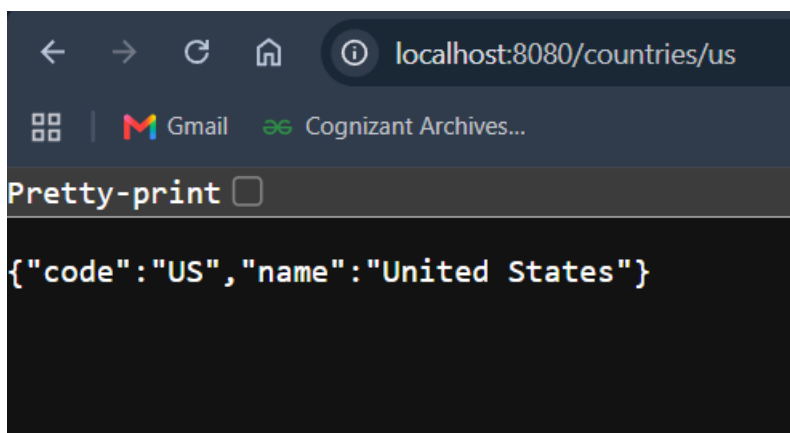
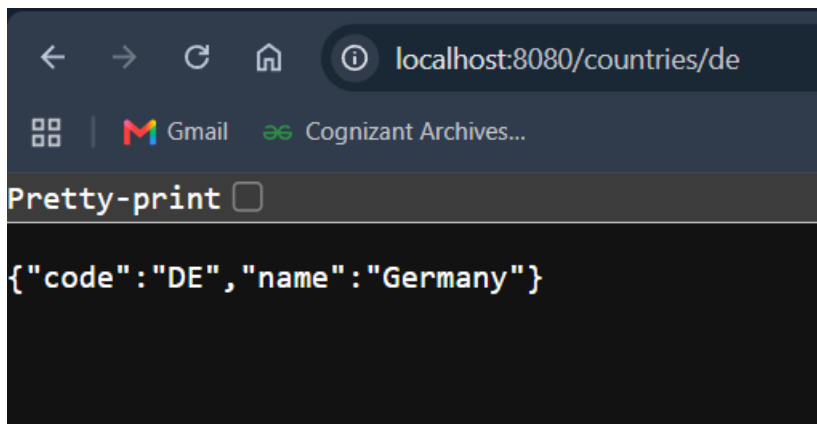
```

earApplication.java    country.xml X    CountryController.java    CountryService.java 2
src > main > resources > country.xml
1  <?xml version="1.0" encoding="UTF-8"?>
2
3  <beans xmlns="http://www.springframework.org/schema/beans"
4      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5      xsi:schemaLocation="
6          http://www.springframework.org/schema/beans
7          https://www.springframework.org/schema/beans/spring-beans.xsd">
8
9      <bean id="countryList" class="java.util.ArrayList">
10         <constructor-arg>
11             <list>
12
13                 <bean class="com.cognizant.springlearn.Country">
14                     <property name="code" value="US"/>
15                     <property name="name" value="United States"/>
16                 </bean>
17
18                 <bean class="com.cognizant.springlearn.Country">
19                     <property name="code" value="DE"/>
20                     <property name="name" value="Germany"/>
21                 </bean>
22
23                 <bean class="com.cognizant.springlearn.Country">
24                     <property name="code" value="IN"/>
25                     <property name="name" value="India"/>
26                 </bean>
27
28                 <bean class="com.cognizant.springlearn.Country">
29                     <property name="code" value="JP"/>
30                     <property name="name" value="Japan"/>
31                 </bean>
32             </list>
33         </constructor-arg>
34     </bean>
35
36 </beans>

```

```
src > main > java > com > cognizant > springlearn > controller > CountryController.java
1 package com.cognizant.springlearn.controller;
2
3 import com.cognizant.springlearn.Country;
4 import com.cognizant.springlearn.service.CountryService;
5 import org.springframework.beans.factory.annotation.Autowired;
6 import org.springframework.web.bind.annotation.*;
7
8 @RestController
9 public class CountryController {
10
11     @Autowired
12     CountryService countryService;
13
14     @GetMapping("/countries/{code}")
15     public Country getCountry(@PathVariable String code) {
16
17         return countryService.getCountry(code);
18     }
19 }
20
21 }
```

RESULT :



```
PS C:\Users\Kanna\OneDrive\Desktop\COGNIZANT\DEEPAK\WEEK1\WEEK_1_MANDATORY\week-4> curl http://localhost:8080/countries/us

StatusCode      : 200
StatusDescription : 
Content         : {"code":"US","name":"United States"}
RawContent      : HTTP/1.1 200
                  Keep-Alive: timeout=60
                  Connection: keep-alive
                  Content-Length: 36
                  Content-Type: application/json
                  Date: Mon, 06 Jul 2026 12:07:59 GMT

                  {"code":"US","name":"United States"}
Forms           : {}
Headers         : [[Keep-Alive, timeout=60], [Connection, keep-alive], [Content-Length, 36],
                  [Content-Type, application/json]...]
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
ResponseHeaders : 36
```

Exercise 6 : Create an authentication service that returns JWT

```
J AuthController.java X
src > main > java > com > cognizant > springlearn > controller > J AuthController.java > AuthCo
1  package com.cognizant.springlearn.controller;
2
3  import com.cognizant.springlearn.util.JwtUtil;
4  import org.springframework.web.bind.annotation.*;
5
6  import java.util.HashMap;
7  import java.util.Map;
8
9  @RestController
10 public class AuthController {
11
12     @PostMapping("/authenticate")
13     public Map<String,String> authenticate(){
14
15         String token = JwtUtil.generateToken(username: "admin");
16
17         Map<String,String> response = new HashMap<>();
18
19         response.put("token", token);
20
21         return response;
22     }
23
24
25 }
```

```
AuthController.java  JwtUtil.java X
src > main > java > com > cognizant > springlearn > util > JwtUtil.java > {} com.cognizant.springlearn.util
1  package com.cognizant.springlearn.util;
2
3  import io.jsonwebtoken.Jwts;
4  import io.jsonwebtoken.SignatureAlgorithm;
5  import io.jsonwebtoken.security.Keys;
6
7  import java.nio.charset.StandardCharsets;
8  import java.security.Key;
9  import java.util.Date;
10
11 public class JwtUtil {
12
13     private static final String SECRET = "mysecretkeymysecretkeymysecretkeymysecretkey";
14     private static final long EXPIRATION_TIME_IN_MILLIS = 600000;
15
16     private JwtUtil() {
17     }
18
19     public static String generateToken(String username) {
20         Key key = Keys.hmacShaKeyFor(SECRET.getBytes(StandardCharsets.UTF_8));
21
22         return Jwts.builder()
23             .setSubject(username)
24             .setIssuedAt(new Date())
25             .setExpiration(new Date(System.currentTimeMillis() + EXPIRATION_TIME_IN_MILLIS))
26             .signWith(key, SignatureAlgorithm.HS256)
27             .compact();
28     }
29 }
```

RESULT :

```
PS C:\Users\Kanna\OneDrive\Desktop\COGNIZANT\DEEPAK\WEEK1\WEEK_1_MANDATORY\week-4> irm -Method POST http://localhost:8083/authenticate
token
-----
eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJhZG1pb2IiLCJhdCI6MTc4MzY0NDQwMCwiZXN1IjozNzgzMzQ1MDAwfQ.5C2Wk_86XjINx1sKgtQttA7K2gv-LhsapJ-f_Ec08

PS C:\Users\Kanna\OneDrive\Desktop\COGNIZANT\DEEPAK\WEEK1\WEEK_1_MANDATORY\week-4> irm -Method POST http://localhost:8083/authenticate
token
-----
eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJhZG1pb2IiLCJhdCI6MTc4MzY0NDQwMCwiZXN1IjozNzgzMzQ1MDAwfQ.Ac40v1MH-ehY-fIbrw2ns2iloYzyWg9LTH_x4Ulvz3g

PS C:\Users\Kanna\OneDrive\Desktop\COGNIZANT\DEEPAK\WEEK1\WEEK_1_MANDATORY\week-4>
```